Computational Complexity

Written Test 1

This test is graded for a maximum of 8.5 points

Skill: *Understanding the big-O notation.*

1. *(1 pts.)* Arrange the following functions in increasing order of growth (i.e. so that a function f on the left of a function g has $f(n) = O(g(n))$):

   (a) n^{10} (b) $(n^2)^{\log n}$ (c) $(1.1)^n$ (d) $(\log n)^{\log n}$ (e) $n^{\log n}$ (f) $100n^2$
   (g) $n!/(2^n)$ (h) $(100)^n$

Skill: *show that a given problem is in* P.

2. *(1 pts.)* Let $G = ([n], E)$ be an undirected graph on n vertices. An *Eulerian tour* of G is a walk in G that traverses every single edge exactly once. I.e., it is a (finite) sequence $(v_0, v_1, \ldots)$ of vertices of G, where vertices can be repeated, but such that every edge of G appears in the sequence exactly once, meaning for every edge $e \in E$ there exists one unique i such that $e = \{v_i, v_{i+1}\}$.

   Show that the problem of deciding whether G has an Eulerian tour can be solved in polynomial time. (*Hint:* the condition of having an Eulerian tour is easy, just try the most obvious, greedy thing. It is very different to the condition of having a Hamiltonian path.)

Skill: *show that a given problem is* NP*-complete.*

3. *(1.5 pts.)* Let

$$\mathsf{Clique} = \{\langle G, k\rangle \mid G \text{ has a clique of size } \geq k\}$$

$$\mathsf{Clique}_{\frac{1}{2}} = \{G \mid G \text{ has a clique of size } \frac{|V(G)|}{2}\}.$$

   We have seen in class that CLIQUEClique is NP-complete. Show that $\mathsf{Clique}_{\frac{1}{2}}$ is also NP-complete.

4. *(2 pts.)* Let G be an undirected graph with vertices $V(G)$ and edges $E(G)$. A *vertex cover* of G is a set $C \subseteq V(G)$ of vertices of G such that every edge $\{u, v\} \in E(G)$ has at least one endpoint (u or v or both) in C. A *dominating set* of G is a set $D \subseteq V(G)$ of vertices of G such that every vertex $v \in V(G) \setminus D$ has at least one neighbour in D (i.e. for all $v \in V(G) \setminus D$ there exists some edge (u, v) with $u \in D$). Then let

$$\mathsf{VertexCover} = \{\langle G, k\rangle \mid G \text{ has a vertex cover of size } \leq k\}$$

$$\mathsf{DominatingSet} = \{\langle G, k\rangle \mid G \text{ has a dominating set of size } \leq k\}.$$

   A homework exercise asked you to show that VertexCover is NP-complete. Now show that DominatingSet is also NP-complete. We suggest that you do this by reduction from VertexCover.

   (A hint for this exercise that gives you the basic gadget will cost you 0.5pts.)

Skill: *Reason about space.*

5. *(1 pts.)* Consider the problem where we are given as input a string $x \in \{), (\}$, i.e. a sequence of parenthesis, and we wish to decide if every parenthesis in x is properly matched. E.g. $(()(()()))$ is properly matched, but $(()))$ is not, because the last closing parenthesis has no corresponding opening parenthesis.

   Show that this problem is in L, i.e., can be solved in logspace.

Skill: *Learn new concepts by yourself.*

Choose 1 out of the following 3 exercises:

6. *(1 pts.)* State and prove the non-deterministic time-hierarchy theorem.

7. *(1.5 pts.)* (Search reduces to decision.) Show that, if there exists a polynomial-time algorithm A to decide whether a given 3CNF is satisfiable, then there exists a polynomial-time algorithm B which, when given a satisfiable 3CNF ψ, outputs a satisfying assignment x (i.e. such that $\psi(x) = 1$).

8. *(2 pts.)* Prove that $\mathsf{NL} = \mathsf{coNL}$.